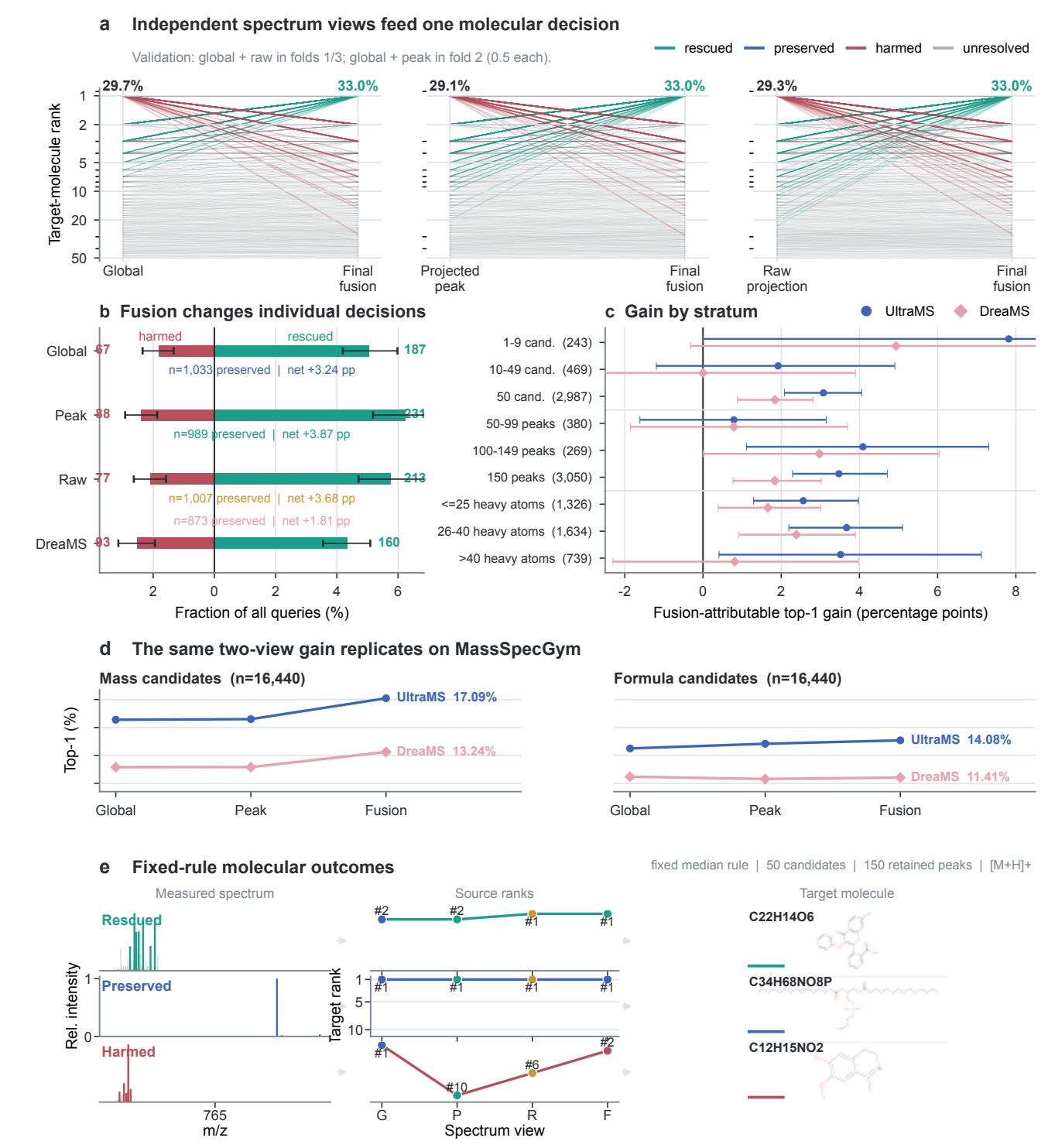

a Independent spectrum views feed one molecular decision
Validation: global + raw in folds 1/3; global + peak in fold 2 (0.5 each).
rescued
preserved
harmed
unresolved
Target-molecule rank
29.7%
33.0%
29.1%
33.0%
29.3%
33.0%
Global
Final fusion
Projected peak
Raw projection
b Fusion changes individual decisions
harmed
rescued
Global
Peak
Raw
DreaMS
n=1,033 preserved | net +3.24 pp
n=989 preserved | net +3.87 pp
n=1,007 preserved | net +3.68 pp
n=873 preserved | net +1.81 pp
Fraction of all queries (%)
c Gain by stratum
UltraMS
DreaMS
1-9 cand. (243)
10-49 cand. (469)
50 cand. (2,987)
50-99 peaks (380)
100-149 peaks (269)
150 peaks (3,050)
<=25 heavy atoms (1,326)
26-40 heavy atoms (1,634)
>40 heavy atoms (739)
Fusion-attributable top-1 gain (percentage points)
d The same two-view gain replicates on MassSpecGym
Mass candidates (n=16,440)
UltraMS 17.09%
DreaMS 13.24%
Top-1 (%)
Global
Peak
Fusion
Formula candidates (n=16,440)
UltraMS 14.08%
DreaMS 11.41%
e Fixed-rule molecular outcomes
fixed median rule | 50 candidates | 150 retained peaks | [M+H]+
Measured spectrum
Source ranks
Target molecule
Rescued
Preserved
Harmed
Rel. intensity
m/z
Target rank
Spectrum view
C22H14O6
C34H68NO8P
C12H15NO2